Project 1: Vending Machine – Design and Verification Report

Name of Team members -:

Course – Computer Organization and Architecture

1. Objective

The objective of this project is to design and implement, in Logisim-evolution, a vending machine controller for a single product costing **Rs. 15**. The machine accepts Rs. 5 and Rs. 10 coin requests, accumulates credit, dispenses the product automatically on sufficient payment, displays change, and refunds undispensed credit on cancellation. Product stock and change reserves are assumed to be unlimited.

2. Interface

All inputs are press-event requests. An input that is ignored does not represent accepted money.

Signal	Direction	Bits	Meaning
CLK	Input	1	Master clock (one period = 1/64 second in the specified timing)
COIN5, COIN10	Input	1 each	Requests to accept a Rs. 5 or Rs. 10 coin
CANCEL	Input	1	Cancel the transaction
RESET	Input	1	Resets the machine to its initial state
CREDIT	Output	5	Accepted credit: 0, 5, 10, 15 or 20 rupees
RETURN_AMOUNT	Output	4	Change or refund, 0–10 rupees
DISPENSE, REFUND	Output	1 each	Product-delivery and refund indicators
BUSY	Output	1	High while a vend or refund is in progress
ERROR	Output	1	High after a conflicting (simultaneous) coin request

CREDIT and RETURN_AMOUNT are each also shown on a separate two-digit display (tens and ones digits). Reset establishes the COLLECT state with CREDIT = 00, RETURN_AMOUNT = 00 and all indicators low.

3. Required Behaviour

The controller has three operating modes.

COLLECT. One coin event is accepted per clock edge and its value is added to the credit. If both coin events occur on the same edge, both are rejected: credit and return amount are retained, ERROR is set to 1, and the machine stays in COLLECT. If the updated credit is below Rs. 15, it is retained and an accepted single coin clears ERROR and the previous RETURN_AMOUNT. If the updated credit reaches Rs. 15 or Rs. 20, the machine enters VEND on that same edge, with RETURN_AMOUNT = credit − 15 and CREDIT retained at 15 or 20.

VEND. DISPENSE = 1 and BUSY = 1 for exactly 64 master periods with REFUND = 0. The machine then returns to COLLECT and clears CREDIT, while RETURN_AMOUNT is retained until another single coin is accepted or reset occurs. One purchase produces one continuous DISPENSE interval.

RETURN. Entered by CANCEL when credit is positive. REFUND = 1 and BUSY = 1 for 64 periods with DISPENSE = 0, then the machine returns to COLLECT and retains RETURN_AMOUNT.

Cancellation. In COLLECT, CANCEL takes priority over either or both coin events and clears ERROR. With positive credit, the credit is copied to RETURN_AMOUNT, CREDIT is cleared immediately and the machine enters RETURN. With zero credit it remains in COLLECT, retains the previous RETURN_AMOUNT and does not start a refund interval.

All coin and CANCEL events are ignored in VEND and RETURN, including on their expiry edges. RESET overrides every state and terminates all activity. BUSY equals DISPENSE OR REFUND, and the two activity indicators are never high together.

4. Design Approach

The controller is implemented as a **ROM-based finite state machine** (a Moore machine whose outputs depend only on the stored state). The design has no discrete logic gates for control: the next-state function, the output decoding and the timing are stored in three read-only memories, and three registers hold the machine's memory. This keeps the circuit compact and makes every specified behaviour a table entry that can be inspected and verified exhaustively.

4.1 Block overview

Block	Type	Function
PREV	3-bit register	Stores the previous value of INB = {CANCEL, COIN10, COIN5} for edge detection
STATE	8-bit register	Holds the current controller state
CNT	6-bit register	Holds the 64-period timer count
ROM_CTRL	ROM, 16-bit address, 8-bit data	Next-state function
ROM_OUT	ROM, 8-bit address, 29-bit	Decodes STATE into outputs and display digits

Block	Type	Function
	data	
ROM_TIMER	ROM, 8-bit address, 7-bit data	Next count value and the terminal-count flag T63

All three registers share the same clock and update on the rising edge.

4.2 Coin and cancel events by edge detection

A coin or cancel event is defined as a **rising edge** of the input, detected by comparing the current input INB with its previous value PREV. This means a button that is held down is counted once, and a held input cannot cause repeated coin acceptance. The previous-input register is updated on every clock edge, including during VEND and RETURN, so an input that is still held when a vend or refund ends does not produce a new event.

4.3 State encoding

The 8-bit state is organised as follows.

Bits	Field	Values
7:6	Mode	00 = COLLECT, 01 = VEND, 10 = RETURN
5:3	Credit index	0–4 for credit 0, 5, 10, 15, 20
2:1	Return index	0–3 for return amount 0, 5, 10, 15
0	Error flag	1 after a conflicting coin request

Because credit, return amount, mode and error are all part of the state, the outputs are a pure function of the state and can be read directly from ROM_OUT.

4.4 Next-state ROM (ROM_CTRL)

The address of ROM_CTRL is formed by combining eight signals into 16 bits:

{ STATE[7:0], RESET, T63, PREV[2:0], INB[2:0] }

The 8-bit data output is the next state, which is loaded into the STATE register on the next clock edge. The table encodes the following rules:

- RESET = 1 forces the next state to 0 regardless of every other input.
- In COLLECT, CANCEL (rising edge) has priority. With credit it produces the RETURN state with RETURN_AMOUNT equal to the credit and CREDIT cleared. Without credit it keeps the state and clears ERROR.
- In COLLECT, a rising edge on both coin inputs gives the same state with the error flag set.
- In COLLECT, a single coin rising edge computes the new credit. Below Rs. 15 it stays in COLLECT with the return amount cleared and error cleared. At Rs. 15 or Rs. 20 it enters VEND with return amount = credit − 15.

- In VEND and RETURN, all coin and CANCEL inputs are ignored. The state is held until T63 is high, after which the machine returns to COLLECT (credit cleared after VEND, return amount retained).

4.5 Timer (ROM_TIMER and CNT)

ROM_TIMER is addressed by {STATE[7:6], CNT[5:0]}. In COLLECT the next count is always 0, so the counter is zero whenever a vend or refund begins. In VEND and RETURN the count increments by one each period. When the count equals 63, T63 goes high and the next count wraps to 0. The state is entered with count 0, so counts 0 to 63 give exactly **64 master periods** of DISPENSE (or REFUND) and BUSY before the machine leaves the state. Because the timer table is addressed using the current mode, a reset in the middle of a vend leaves no stale count that could shorten the next vend.

4.6 Output ROM (ROM_OUT)

ROM_OUT is addressed by the STATE register. Its 29-bit output word is split as follows:

Bits	Signal
4:0	CREDIT
8:5	RETURN_AMOUNT
9	DISPENSE
10	REFUND
11	BUSY
12	ERROR
16:13	Credit tens digit
20:17	Credit ones digit
24:21	Return amount tens digit
28:25	Return amount ones digit

BUSY is stored as 1 exactly for the VEND and RETURN modes, so BUSY = DISPENSE OR REFUND holds by construction, and DISPENSE and REFUND are never 1 in the same state. The digit fields drive four hex digit displays so that the displays always agree with the numeric outputs.

4.7 Reset

RESET is part of the ROM_CTRL address. When it is high at a clock edge, the next state is 0, which decodes to CREDIT = 0, RETURN_AMOUNT = 0, ERROR = 0, DISPENSE = 0, REFUND = 0 and BUSY = 0 in the COLLECT mode. Reset therefore overrides every state, including VEND and RETURN, and terminates any activity.

5. Verification

5.1 Method

The design was verified in two ways. First, the circuit file was analysed directly: the wiring of the registers, ROMs, splitters, tunnels and displays was traced, and the contents of all three ROMs were decoded. Second, an independent reference model was written from the problem statement alone and compared with the decoded design. Both machines were driven with all 16 combinations of COIN5, COIN10, CANCEL and RESET from every state reachable after reset (including the timer count and previous-input values), and CREDIT, RETURN_AMOUNT, DISPENSE, REFUND, BUSY, ERROR and the display digits were compared after every clock edge. In total, **2,618 reachable combined states were explored with 0 mismatches.**

The same exhaustive run also confirmed that BUSY equals DISPENSE OR REFUND in every state, that DISPENSE and REFUND are never high together, and that the two-digit displays always equal CREDIT and RETURN_AMOUNT. In addition, RESET was checked from all 256 possible states and previous-input and count values, and it always leads to the all-zero state.

5.2 Directed test scenarios

The verification checks required by the problem statement were also run as explicit scenarios. In each case, a period is one clock cycle.

Scenario	Observed result
Exact payment (5 + 5 + 5)	Credit 5, 10, then 15. VEND starts on the third coin with CREDIT = 15 and RETURN_AMOUNT = 0. DISPENSE and BUSY are high for 64 periods, then CREDIT returns to 0.
Overpayment (10 + 10)	VEND starts on the second coin with CREDIT = 20 and RETURN_AMOUNT = 5. DISPENSE and BUSY are high for 64 periods. CREDIT clears, RETURN_AMOUNT stays at 5.
Cancel with partial credit (10)	CREDIT clears immediately and RETURN_AMOUNT = 10. REFUND and BUSY are high for 64 periods with DISPENSE = 0, then RETURN_AMOUNT is retained.
Simultaneous coins	Credit and return amount are unchanged and ERROR = 1. A later CANCEL clears ERROR and starts a refund of the existing credit.
CANCEL together with a coin	CANCEL takes priority: the coin is not added and the existing credit is refunded.
Held input (COIN5 held for 10 cycles)	Credit increases by 5 once only.
Coin and CANCEL activity during VEND	All ignored. DISPENSE stays high for exactly 64

Scenario	Observed result
	periods, one continuous interval, and no extra credit is added.
Coin or CANCEL exactly on the VEND or RETURN expiry edge	Ignored. The machine returns to COLLECT with the retained RETURN_AMOUNT and no spurious credit or refund.
CANCEL with zero credit	The machine stays in COLLECT, the previous RETURN_AMOUNT is retained and no refund interval starts.
RESET during VEND	All outputs return to zero on the next clock edge and the activity ends.

5.3 Results

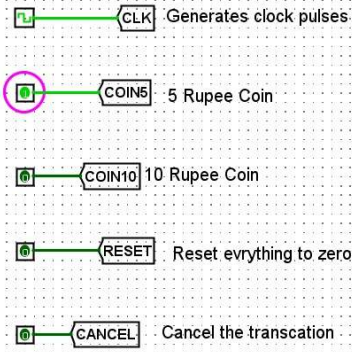
All scenarios and the exhaustive comparison matched the specification. The DISPENSE and REFUND intervals are exactly 64 periods in length, the two activity indicators are never high together, and BUSY is always consistent with them.

6. Assumptions

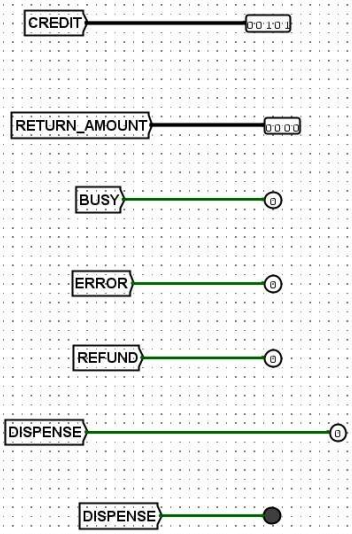
The problem statement leaves two points open, and the design follows the readings below.

- A coin or CANCEL "event" is a rising edge of the input rather than a level sampled at every clock edge. This is the reading under which a held input is counted only once.
- ERROR is sticky: once set by a conflicting coin request it stays high until the next accepted single coin, a CANCEL or a RESET, and it is not cleared by idle clock cycles.
- RESET acts synchronously, through the next-state ROM, so it takes effect at the next clock edge.

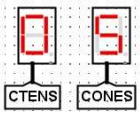
Inputs



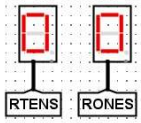
Outputs

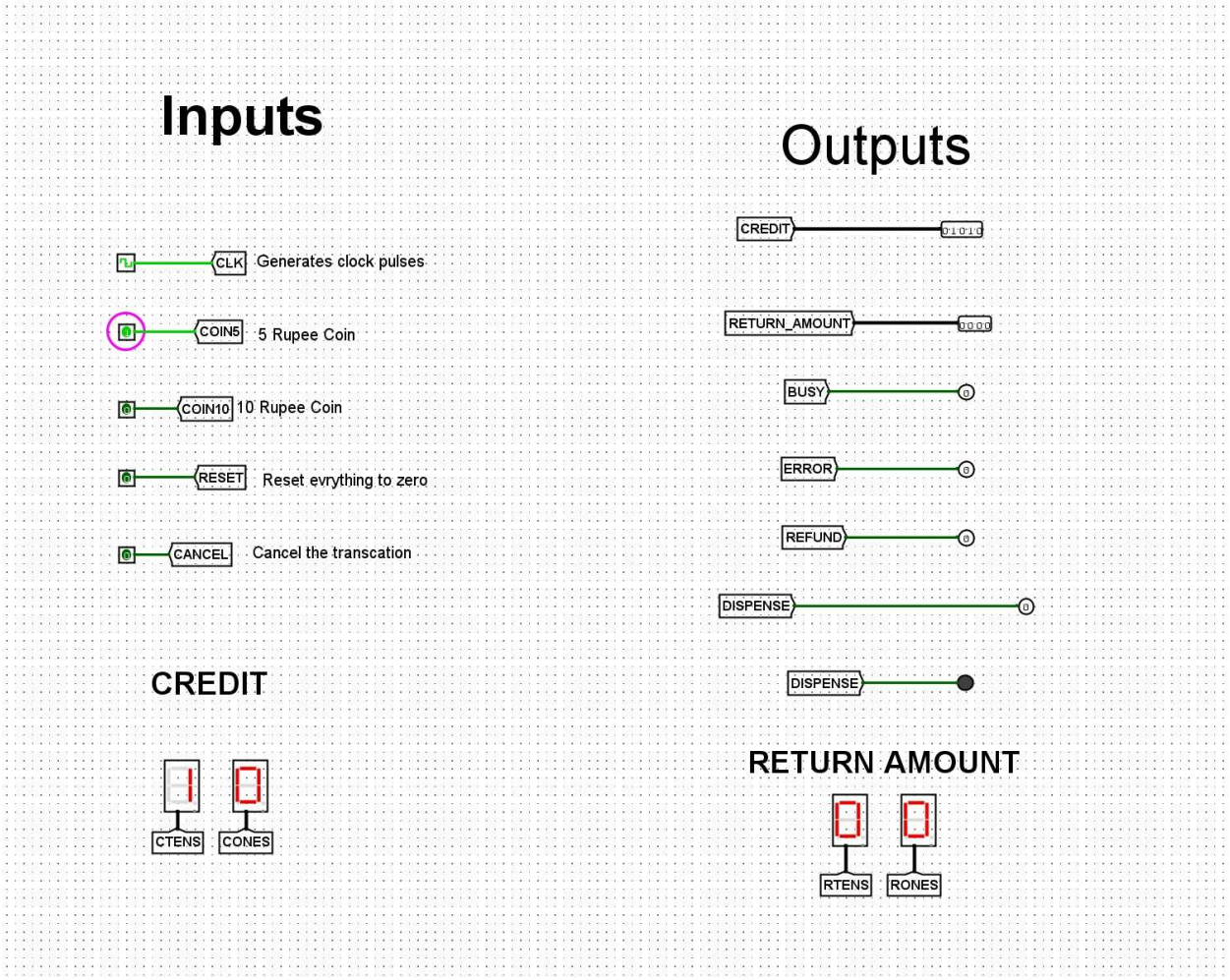


CREDIT

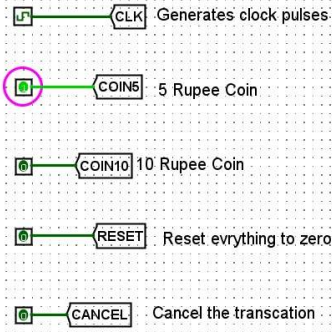


RETURN AMOUNT

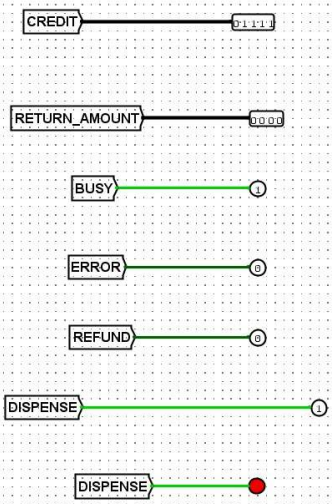



Inputs
Outputs
CLK
Generates clock pulses
COIN5
5 Rupee Coin
COIN10
10 Rupee Coin
RESET
Reset evrything to zero
CANCEL
Cancel the transcation
CREDIT
RETURN_AMOUNT
BUSY
ERROR
REFUND
DISPENSE
DISPENSE
CREDIT
CTENS
CONES
RETURN AMOUNT
RTENS
RONES

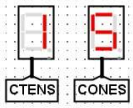
Inputs



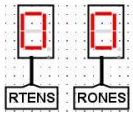
Outputs



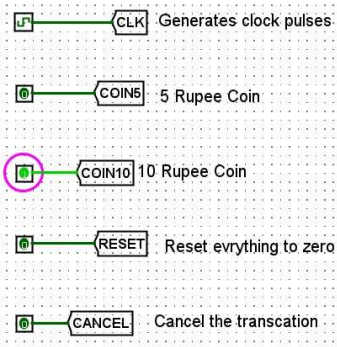
CREDIT



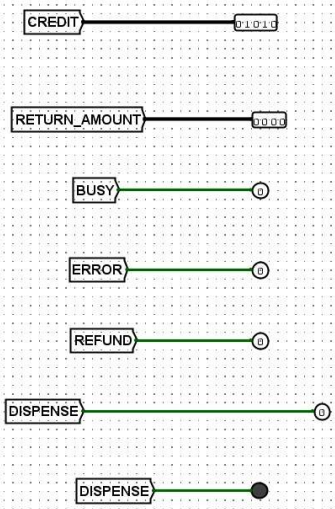
RETURN AMOUNT



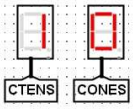
Inputs



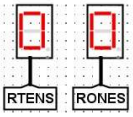
Outputs

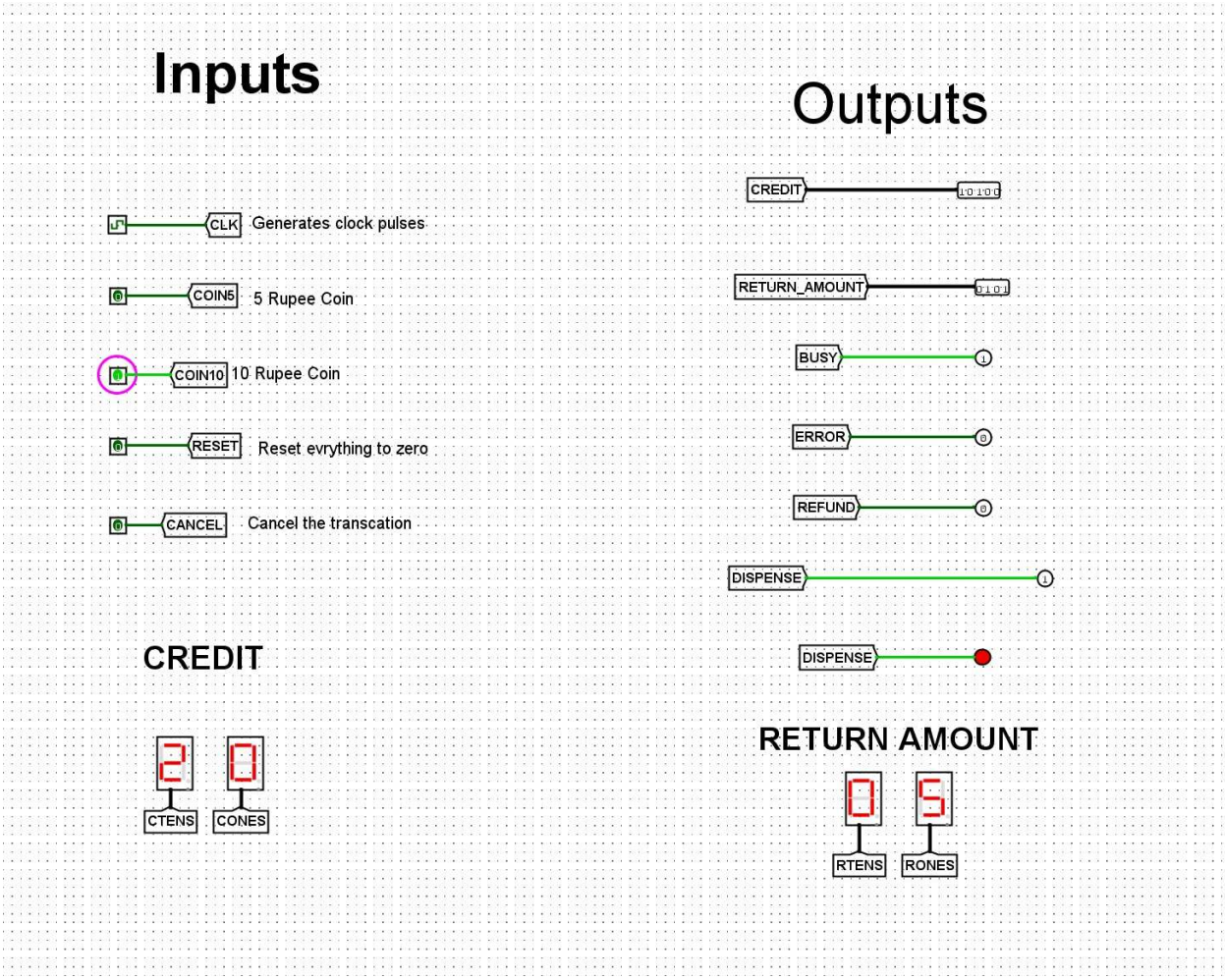


CREDIT

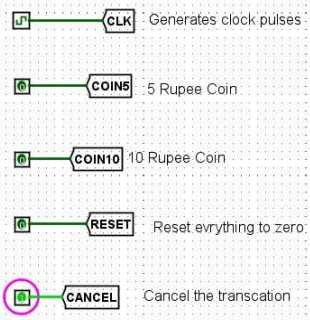


RETURN AMOUNT

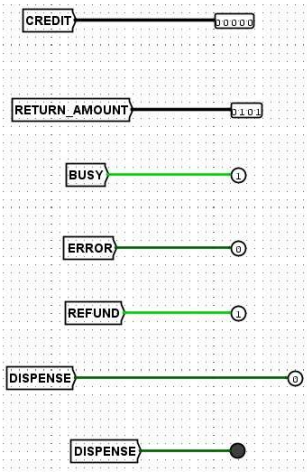



Inputs
CLK
Generates clock pulses
COIN5
5 Rupee Coin
COIN10
10 Rupee Coin
RESET
Reset evrything to zero
CANCEL
Cancel the transcation
Outputs
CREDIT
RETURN_AMOUNT
BUSY
ERROR
REFUND
DISPENSE
DISPENSE
CREDIT
CTENS
CONES
RETURN AMOUNT
RTENS
RONES

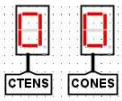
Inputs



Outputs



CREDIT



RETURN AMOUNT

